

AI-Assisted Heart Disease Prediction Project

AI-Assisted Machine Learning Final Exam Submission
Research Name: AI-Assisted Heart Disease Prediction Project
Application: Heart Disease Prediction System

Date	July 8, 2026
Course	Artificial Intelligence Final Examination
Research Name	AI-Assisted Heart Disease Prediction Project
Live Website	https://heartdiseaseaimllabfinal.streamlit.app/
Best Model	Random Forest
Best Accuracy	93.44%

To
The Course Instructor
Department of Computer Science / Artificial Intelligence

Subject: Submission of final AI-assisted machine learning project report.

Dear Sir/Madam,

I am submitting my final examination project titled **AI-Assisted Heart Disease Prediction Project**. The project focuses on building a machine learning system that predicts whether a patient may have heart disease using the UCI Cleveland Heart Disease dataset.

The submission includes dataset understanding, exploratory data analysis, preprocessing, feature engineering, model training, model evaluation, model interpretation, AI-assisted development reflection, ethical discussion, and deployment details. Three machine learning models were compared: Logistic Regression, Random Forest, and Support Vector Machine. Based on the evaluation results, Random Forest was selected as the best model.

A Streamlit web application was also created so users can test single-patient and multi-patient predictions interactively. The live app link is included in this report, along with a screenshot and explanation of the main website functionality.

I have kept the implementation clear and suitable for an academic final exam project. The prediction system is for educational use only and should not be used as a real medical diagnosis tool without clinical validation.

Sincerely,

Submitted by: _____

Student ID: _____

Section/Batch: _____